

HDOC DAY – 21

Question:

Write a menu-driven C program that accepts two positive integers, m and n , where $m \leq n$. Display the following menu and accept the user's choice. Use a switch statement to perform the selected operation:

- 1. Print the first 10 multiples of each integer from m to n .*
- 2. Print the addition table of each integer from m to n , using addends from 1 to 10.*
- 3. Print the subtraction table of each integer from m to n , using subtrahends from 1 to 10.*
- 4. Print the results obtained by dividing each integer from m to n by the integers from 1 to 10. Display the results rounded to two decimal places.*
- 5. Exit the program.*

Display an appropriate error message if the user enters an invalid menu choice. Continue displaying the menu and performing the selected operation until the user chooses the Exit option.

Note: Prepare the algorithm and test-case table. Then write, compile and execute the C program, and test it using the prepared test cases.

Algorithm:

Step 1: Start.

Step 2: Declare integer variables m , n , choice, i , and j .

Step 3: Input two positive integers m and n .

Step 4: Check whether the input is valid:

- m and n must be positive.

- $m \leq n$.
- If invalid, display an error message and ask for m and n again.

Step 5: Display the menu:

Step 6: Accept the user's choice.

Step 7: Use a switch statement.

- Case 1: For every number from m to n, print its first 10 multiples.
- Case 2: For every number from m to n, add the numbers 1 to 10 and display the results.
- Case 3: For every number from m to n, subtract the numbers 1 to 10 and display the results.
- Case 4: For every number from m to n, divide it by the numbers 1 to 10 and display the result up to two decimal places.
- Case 5: Display an exit message and terminate the program.
- Default: Display "Invalid menu choice".

Step 8: Repeat Steps 5–7 until the user selects 5.

Step 9: Stop.

C program:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int m, n, choice, i, j;
```

```
    /* Input m and n */
```

```
    printf("Enter two positive integers m and n (m <= n): ");
```

```
scanf("%d %d", &m, &n);
```

```
/* Input validation */
```

```
while (m <= 0 || n <= 0 || m > n)
```

```
{
```

```
    printf("Invalid input! Enter positive integers with m <= n.\n");
```

```
    printf("Enter m and n again: ");
```

```
    scanf("%d %d", &m, &n);
```

```
}
```

```
/* Menu will repeat until user chooses Exit */
```

```
do
```

```
{
```

```
    printf("\n----- MENU ----- \n");
```

```
    printf("1. Multiplication Table\n");
```

```
    printf("2. Addition Table\n");
```

```
    printf("3. Subtraction Table\n");
```

```
    printf("4. Division Table\n");
```

```
    printf("5. Exit\n");
```

```
    printf("Enter your choice: ");
```

```
    scanf("%d", &choice);
```

```
    switch (choice)
```

```
{
```

case 1:

```
/* Multiplication tables */  
for (i = m; i <= n; i++)  
{  
    printf("\nMultiplication table of %d:\n", i);  
  
    for (j = 1; j <= 10; j++)  
    {  
        printf("%d x %d = %d\n", i, j, i * j);  
    }  
}  
break;
```

case 2:

```
/* Addition tables */  
for (i = m; i <= n; i++)  
{  
    printf("\nAddition table of %d:\n", i);  
  
    for (j = 1; j <= 10; j++)  
    {  
        printf("%d + %d = %d\n", i, j, i + j);  
    }  
}  
break;
```

case 3:

```
/* Subtraction tables */  
for (i = m; i <= n; i++)  
{  
    printf("\nSubtraction table of %d:\n", i);  
  
    for (j = 1; j <= 10; j++)  
    {  
        printf("%d - %d = %d\n", i, j, i - j);  
    }  
}  
break;
```

case 4:

```
/* Division tables */  
for (i = m; i <= n; i++)  
{  
    printf("\nDivision table of %d:\n", i);  
  
    for (j = 1; j <= 10; j++)  
    {  
        printf("%d / %d = %.2f\n",  
            i, j, (float)i / j);  
    }  
}
```

```
    }  
    break;  
  
case 5:  
    printf("\nExiting the program...\n");  
    break;  
  
default:  
    printf("\nInvalid menu choice! Please enter 1 to 5.\n");  
}  
  
} while (choice != 5);  
  
return 0;  
}
```

```

3 ~ 6 ~ 1 ~ 2
3 ~ 7 ~ 1 ~ 3
3 ~ 8 ~ 1 ~ 4
3 ~ 9 ~ 1 ~ 5
3 ~ 10 ~ 1 ~ 6

Subtraction table:
3 ~ 1 ~ 3 ~ 2
3 ~ 2 ~ 3 ~ 1
3 ~ 3 ~ 3 ~ 0
3 ~ 4 ~ 3 ~ -1
3 ~ 5 ~ 3 ~ -2

case 2:
/* Addition tables */
for (i = m; i ≤ n; i++)
{
    printf("\nAddition table of %d:\n", i);

    for (j = 1; j ≤ 10; j++)
    {
        printf("%d + %d = %d\n", i, j, i + j);
    }
}
break;

3 ~ 6 ~ 3 ~ 1
3 ~ 7 ~ 3 ~ 2
3 ~ 8 ~ 3 ~ 3
3 ~ 9 ~ 3 ~ 4
3 ~ 10 ~ 3 ~ 5

case 3:
/* Subtraction tables */
for (i = m; i ≤ n; i++)
{
    printf("\nSubtraction table of %d:\n", i);

    for (j = 1; j ≤ 10; j++)
    {
        printf("%d - %d = %d\n", i, j, i - j);
    }
}
break;

----- MENU -----
1. Multiplication Table
2. Addition Table
3. Subtraction Table
4. Division Table
5. Exit
Enter your choice: 3
Subtraction table:
case 4:
/* Division tables */
for (i = m; i ≤ n; i++)
{
    printf("\nDivision table of %d:\n", i);

    for (j = 1; j ≤ 10; j++)
    {
        printf("%d / %d = %.2f\n",
            i, j, (float)i / j);
    }
}
break;

Subtraction table of 3:
case 5:
printf("\nExiting the program... \n");
break;

default:
printf("\nInvalid menu choice! Please enter 1 to 5.\n");
}
} while (choice ≠ 5);

return 0;
}

----- MENU -----

```

```

Session Actions Edit View Help
#include <stdio.h>
Enter your choice:
int main()
{
    int m, n, choice, i, j;

    /* Input m and n */
    printf("Enter two positive integers m and n (m ≤ n): ");
    scanf("%d %d", &m, &n);

    /* Input validation */
    while (m ≤ 0 || n ≤ 0 || m > n)
    {
        printf("Invalid input! Enter positive integers with m ≤ n.\n");
        printf("Enter m and n again: ");
        scanf("%d %d", &m, &n);
    }

    /* Menu will repeat until user chooses Exit */
    do
    {
        printf("\n—— MENU ——\n");
        printf("1. Multiplication Table\n");
        printf("2. Addition Table\n");
        printf("3. Subtraction Table\n");
        printf("4. Division Table\n");
        printf("5. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                /* Multiplication tables */
                for (i = m; i ≤ n; i++)
                {
                    printf("\nMultiplication table of %d:\n", i);
                    for (j = 1; j ≤ 10; j++)
                    {
                        printf("%d x %d = %d\n", i, j, i * j);
                    }
                }
                break;

            case 2:
                /* Addition tables */
                for (i = m; i ≤ n; i++)
                {
                    printf("\nAddition table of %d:\n", i);
                    for (j = 1; j ≤ 10; j++)

```

Compilation and execution:

```

(kali㉿kali)-[~]
$ vi 21.c

(kali㉿kali)-[~]
$ gcc 21.c -o 21

(kali㉿kali)-[~]
$ ./21
Enter two positive integers m and n (m ≤ n): 2 3

```

Test case table:

Test case	m	n	choice	type	Expected result
1	2	3	1	valid	Prints first 10 multiples of 2 and 3
2	2	3	2	Valid	Prints addition tables of 2 and 3 using 1-10
3	5	6	3	Valid	Prints subtraction tables of 5 and 6 using 1-10
4	2	3	4	Valid	Prints division tables of 2 and 3 up to 2 decimal places
5	1	1	1	Boundary	Prints first 10 multiples of 1
6	0	5	--	Invalid	Displays error because m is not positive
7	2	-4	--	Invalid	Displays error because n is not positive
8	8	4	--	Invalid	Displays error because $m > n$
9	2	5	7	Invalid	Displays "Invalid menu choice"
10	2	5	5	Valid	Exits the program